

IInd SEMESTER B. Tech. (EE)

MID-SEMESTER EXAMINATION, FEBRUARY- 2026

Course Code: EEEEC03

Course Title: Signal and

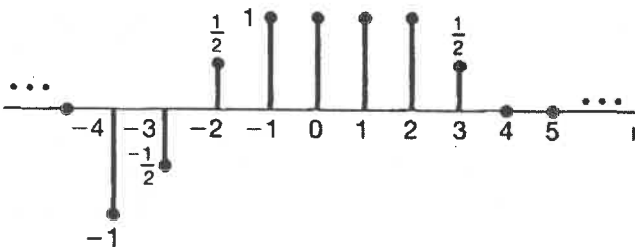
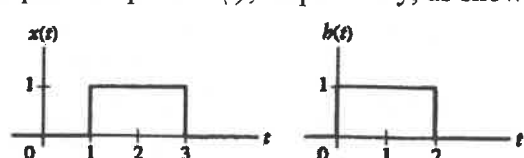
System

Time: 1.5 Hours

Max.

Marks: 25

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q No	Question	Marks	CO
1.a.	Determine the values of power and energy for the following signal: $x(t) = \cos(2t)$	2	CO1
1.b.	A discrete-time signal $x[n]$ is shown as follows.  Sketch and label carefully each of the following signals: (i) $x[n-4]$ (ii) $x[3n+1]$ (iii) $x[n-2]\delta[n-2]$	3	CO1
2.a.	Find the fundamental time period of the signal $y(t) = 10 \cos\left(12t + \frac{\pi}{2}\right) - 3 \sin\left(5t - \frac{\pi}{2}\right)$	2	CO1
2.b.	Check whether the following system is LTI or not. $y(t) = \begin{cases} 0, & t < 0 \\ x(t) + x(t-2), & t \geq 0 \end{cases}$	3	CO1
3.a.	An LTI system has impulse response $h(t) = e^{-2t}$. What should be the output of the system if it is subjected to an input of a delayed unit step signal $u(t-3)$?	2	CO1
3.b.	Evaluate the convolution integral for a system with input $x(t)$ and impulse response $h(t)$, respectively, as shown below. 	3	CO2

4.a.	Determine the Laplace Transform of the following signal $x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t)$. Show the ROC	2	CO2
4.b.	Evaluate the Laplace Transform of the following signal $f(t) = \left(\frac{1-e^{-5t}}{t}\right)u(t)$	3	CO2
5.a.	Find even and odd component of $x(t) = e^{\sigma t}$	2	CO1
5.b.	Evaluate the bilateral Laplace Transform of the following signal $f(t) = e^{at} \sin \omega t$	3	CO2